

TF-IDF Vector Representation for Each Document

1. Preprocessing (Stopword Removal)

Given the stopwords list: `and, be, is, it, to, will`, we remove them from each document:

- **Document 1:** "It is going to rain and rain and rain today."
→ `["going", "rain", "rain", "rain", "today"]`
- **Document 2:** "Today I will be playing sport."
→ `["I", "playing", "sport", "today"]`
(Note: "I" is retained as it's not in the stoplist.)
- **Document 3:** "I am going to watch the play."
→ `["I", "am", "going", "watch", "play"]`
(Note: "am" is retained as it's not in the stoplist.)

2. Vocabulary (Unique Terms)

After processing, the full vocabulary (sorted alphabetically) is:

`["am", "going", "I", "play", "playing", "rain", "sport", "today", "watch"]`

3. Term Frequency (TF) Calculation

TF for term `t` in document `d` is:

$$TF(t, d) = \frac{\text{Number of times } t \text{ appears in } d}{\text{Total terms in } d}$$

Term	Doc1 (5 terms)	Doc2 (4 terms)	Doc3 (5 terms)
am	0	0	1/5 = 0.2
going	1/5 = 0.2	0	1/5 = 0.2
I	0	1/4 = 0.25	1/5 = 0.2
play	0	0	1/5 = 0.2
playing	0	1/4 = 0.25	0
rain	3/5 = 0.6	0	0
sport	0	1/4 = 0.25	0
today	1/5 = 0.2	1/4 = 0.25	0
watch	0	0	1/5 = 0.2

4. Inverse Document Frequency (IDF) Calculation

IDF for term t is:

$$IDF(t) = \log \left(\frac{\text{Total documents } N}{\text{Documents containing } t} \right) \text{ Here, } (N = 3).$$

Term	Docs Containing Term	IDF = $\log(3/df)$
am	1 (Doc3)	$\log(3/1) \approx 1.099$
going	2 (Doc1, Doc3)	$\log(3/2) \approx 0.405$
I	2 (Doc2, Doc3)	$\log(3/2) \approx 0.405$
play	1 (Doc3)	$\log(3/1) \approx 1.099$
playing	1 (Doc2)	$\log(3/1) \approx 1.099$
rain	1 (Doc1)	$\log(3/1) \approx 1.099$
sport	1 (Doc2)	$\log(3/1) \approx 1.099$
today	2 (Doc1, Doc2)	$\log(3/2) \approx 0.405$
watch	1 (Doc3)	$\log(3/1) \approx 1.099$

5. TF-IDF Vectors

Multiply TF and IDF for each term in each document:

[
TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)
]

Term	Doc1	Doc2	Doc3
am	0	0	$0.2 \times 1.099 \approx 0.220$
going	$0.2 \times 0.405 \approx 0.081$	0	$0.2 \times 0.405 \approx 0.081$
I	0	$0.25 \times 0.405 \approx 0.101$	$0.2 \times 0.405 \approx 0.081$
play	0	0	$0.2 \times 1.099 \approx 0.220$
playing	0	$0.25 \times 1.099 \approx 0.275$	0
rain	$0.6 \times 1.099 \approx 0.659$	0	0
sport	0	$0.25 \times 1.099 \approx 0.275$	0
today	$0.2 \times 0.405 \approx 0.081$	$0.25 \times 0.405 \approx 0.101$	0
watch	0	0	$0.2 \times 1.099 \approx 0.220$

6. Final TF-IDF Vectors

Document 1:

[0, 0.081, 0, 0, 0, 0.659, 0, 0.081, 0]

(Order: am, going, I, play, playing, rain, sport, today, watch)

Document 2:

[0, 0, 0.101, 0, 0.275, 0, 0.275, 0.101, 0]

Document 3:

[0.220, 0.081, 0.081, 0.220, 0, 0, 0, 0, 0.220]

Key Observations

1. **Sparse Vectors:** Most terms do not appear in all documents (many 0s).
2. **High-Weight Terms:**
 - Doc1: "rain" (TF-IDF=0.659) dominates as it appears frequently in only one document.
 - Doc2: "playing" and "sport" have equal weights (both are unique to Doc2).
 - Doc3: "am", "play", and "watch" share the highest weight (all unique to Doc3).
3. **Stopword Impact:** Words like "today" (appearing in 2 docs) receive lower IDF penalties.

This representation allows ranking documents by relevance to a query (e.g., a search for "rain play" would prioritize Doc1 and Doc3).